[INCH-POUND]

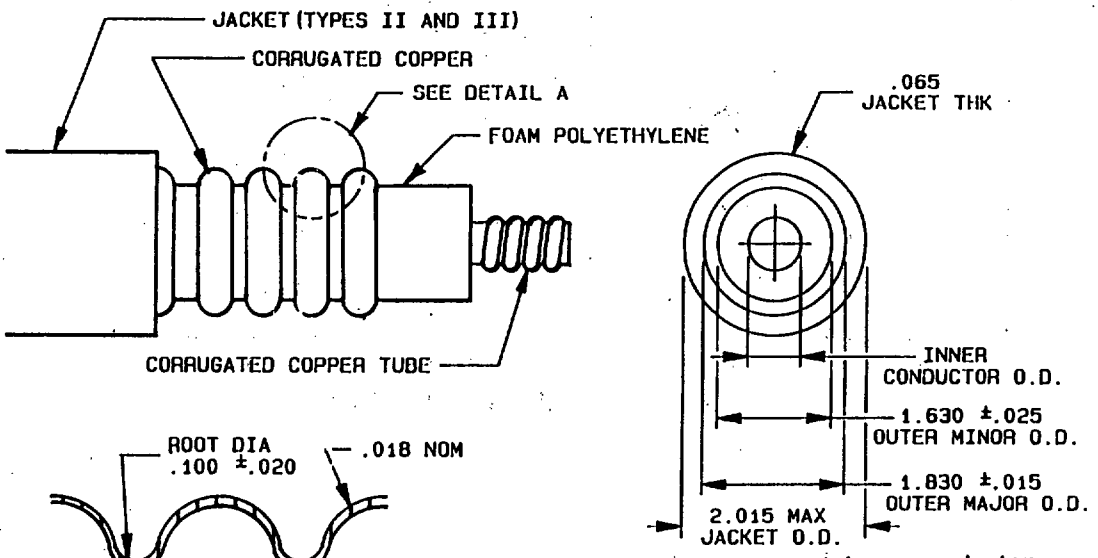
MIL-C-28830/5B
12 June 1989
SUPERSEDING
MIL-C-28830/5A
27 January 1982

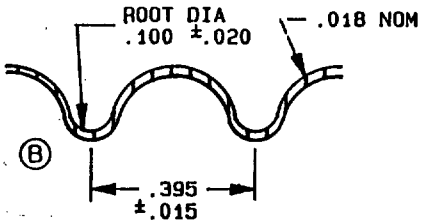
MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, COAXIAL, SEMIRIGID, CORRUGATED OUTER CONDUCTOR, LOW LOSS FOAM DIELECTRIC, 1-5/8 INCH NOMINAL, 50 OHMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-28830.





DETAIL A

* Helically corrugated inner conductor.
Outside diameter: 0.680 ±.005
Strip thickness: .019 nom
Minor outside diameter: .570 ±.015
Pitch: .400 ±.010

Inches	mm	Inches	mm	Inches	mm
.005	0.13	.020	0.51	.570	14.48
.010	0.25	.025	0.64	.680	17.27
.015	0.38	.065	1.65	1.630	41.40
.017	0.43	.100	2.54	1.830	46.48
.018	0.46	.395	10.03	2.015	51.18
.019	0.48	.400	10.16		

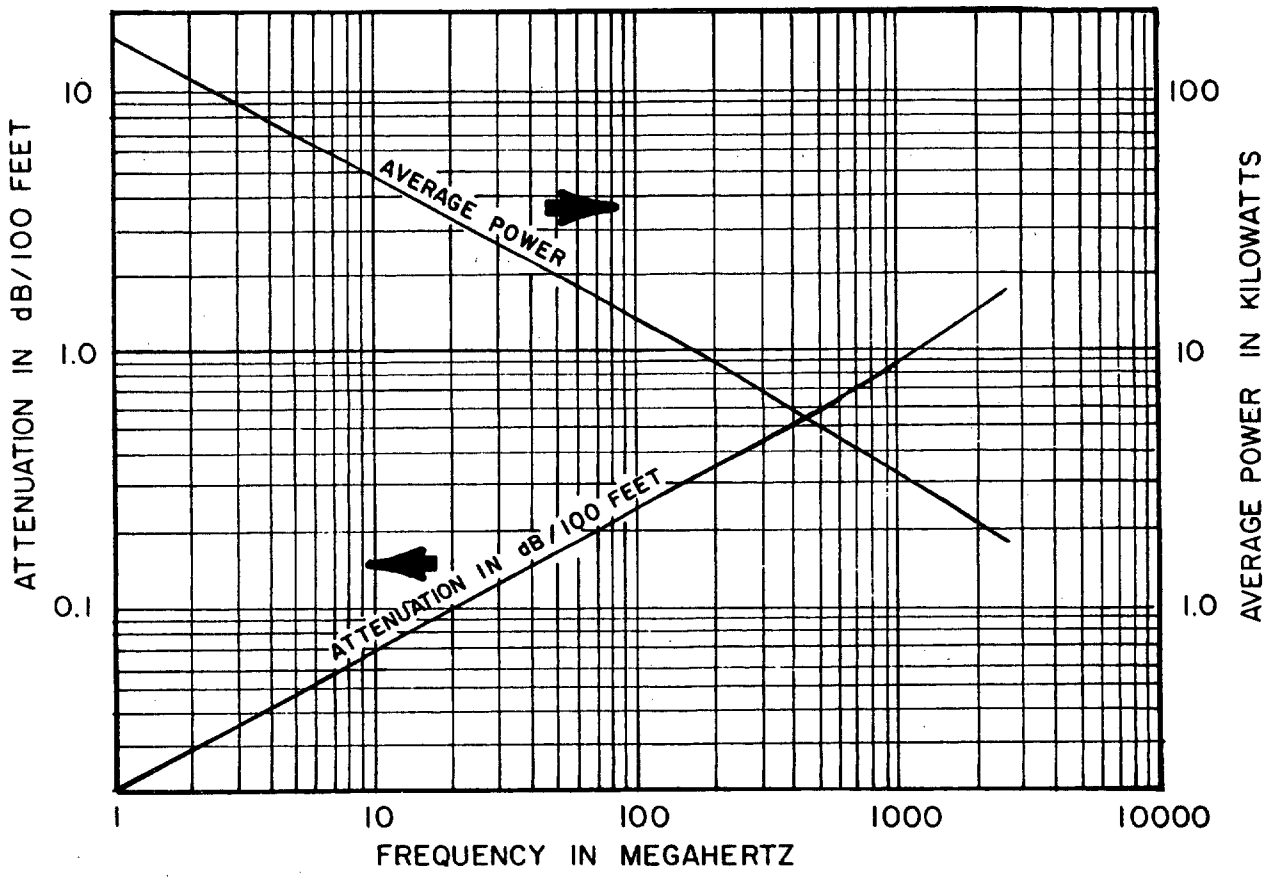
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Cable configuration.

Ⓑ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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STANDARD CONDITIONS

For attenuation
VSWR 1.0
Ambient temperature
24°C (75°F)

For average power
VSWR 1.0
Ambient temperature
40°C (104°F)
Inner conductor
temperature 100°C
(212°F)

FIGURE 2. Attenuation and power curve.

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REQUIREMENTS:

Flexibility and cold bend mandrel diameter: 40 inches (101.6 cm) maximum. (Test cable length: 30 feet (9.144 m).)

Impedance: 50 ±2 ohms.

Attenuation: 0.53 dB per 100 feet (30.48 m) at 400 megahertz (MHz).
0.97 dB per 100 feet (30.48 m) at 1000 MHz.
1.55 dB per 100 feet (30.48 m) at 2000 MHz.

Velocity of propagation: 88 percent nominal.

Capacitance: 23.1 pF per foot nominal.

Voltage standing wave ratio (VSWR): 1.70 maximum.

VSWR frequency range: 100 MHz to 2.4 gigahertz (GHz).

Dielectric withstanding voltage: 15,000 V dc.

Jacket spark: 10,000 V rms.

(B) Part or Identifying Numbers (PINs): M28830/5-II
M28830/5-III

ENGINEERING INFORMATION:

Weight: 1.00 pound per foot nominal.

Minimum in-service bending diameter: 40 inches (101.6 cm).

Average power rating: See figure 2.

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - AS, MC
Air Force - 17, 99
DLA - IS

User activities:
Navy - SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 6145-1140)

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